Simple ALM hedging model: Pacific Life RILA, FIA, and IUL crediting segments

Illustrative model. Valuation 2026-10-26, maturity 2027-10-26, T = 1. Insurer sign convention: positive P&L is a gain to the insurer.

1. Product terms

Product (S&P 500 price return)	Account value	Crediting	Cap	Protection	Fee
Pacific Protective Growth RILA	$100,000	1-year point-to-point, 100% part.	16.25%	10% buffer	none
Pacific Index Advisory FIA	$100,000	1-year point-to-point, 100% part.	10.00%	0% floor	none
Pacific Horizon IUL 2 (1-Year High Cap Indexed Account)	$10,000	1-year point-to-point, 100% part.	12.00%	0% floor	0.80 pts / yr

Market 2026-10-26: 1y Treasury 4.47% quoted (continuous 4.3730%), dividend yield 1.064%, uniform 1y ATMF vol 16.72%. Maturity 2027-10-26, T = 1.

2. Cash flows and embedded options

Insurer receives AV_0 now and owes AV_T = AV_0 x (1 + credited rate) at maturity, where credited rate = option package payoff per $1 - annual deduction. Liability = bond (PV of AV_0) + option (PV of the package) - fee (PV of the deduction); insurer value is its negative.



Embedded option legs	Net notional $	PV $	From
Long calls at 100.00%	+210,000	+17,265	RILA+FIA+IUL
Short calls at 110.00%	-100,000	-4,092	FIA
Short calls at 112.00%	-10,000	-351	IUL
Short calls at 116.25%	-100,000	-2,491	RILA
Short puts at 90.00%	-100,000	-1,762	RILA
Net embedded option package		**+8,569**	**all**

3. Valuation and Greeks (liability view, $)

	RILA	FIA	IUL	Portfolio
Bond PV	95,721	95,721	9,572	201,015
Option PV	3,969	4,129	471	8,569
Fee PV	0	0	77	77
Liability PV = bond + option - fee	99,690	99,850	9,967	209,507
Pricing margin = AV - liability	310	150	33	493
Delta ($ per 1% index)	518.33	221.99	26.21	766.53
Gamma ($ per 1%^2)	-12.38	0.05	0.09	-12.25
Vega ($ per vol pt)	-207.01	0.76	1.45	-204.80
Rho ($ per bp)	-4.79	-7.77	-0.73	-13.29

4. Net exposures (insurer view)

The insurer is short the liability: it loses $767 per 1% index rise, is long gamma (+12 per 1%^2) and long vega (+205 per vol point) because the short cap calls and the RILA buffer put give back more convexity than the long ATM calls provide, and carries +13 per bp of rate exposure, of which only -7 comes from the options; the rest is the $201,015 bond component. Unhedged, the option component swings $28,606 across the stresses below.

5. Hedge recommendations

Buy S&P 500 index futures with $76,653 of index exposure (741.58 units at level 100); no premium, margin and roll excluded. Static alternative: buy the net option package above for $8,569 (= option PV, dealer spread excluded), no rebalancing. Re-hedge the futures when residual index exposure exceeds $4,200 (2% of AV; a 5% index move or a 5-point vol move breaches it), at least monthly, and through the final month to maturity. Manage bond-component rate risk with asset duration.

Full-revaluation stress ($, insurer view)	Bond	Option	Unhedged total	Futures	Option + futures	Hedged total
Index -20%	+0	+16,734	+16,734	-15,331	+1,403	+1,403
Index -10%	+0	+8,120	+8,120	-7,665	+455	+455
Index +10%	+0	-6,900	-6,900	+7,665	+765	+765
Index +20%	+0	-11,873	-11,873	+15,331	+3,458	+3,458
Vol +5 pts	+0	+1,244	+1,244	+0	+1,244	+1,244
Rates +50 bp	+1,003	-338	+664	+384	+46	+1,048
Selloff: index -15%, vol +8, rates -25 bp	-503	+13,365	+12,863	-11,661	+1,705	+1,202
Range (max - min)		**28,606**	**28,606**		**4,122**	**4,505**

6. Explanation of the results

- The RILA's 16.25% cap costs $3,969, less than the FIA's $4,129 for a 10% cap, because the policyholder has written a 90% put worth $1,762 that funds a further-out cap call.
- The $8,985 interest budget on $210,000 pays for $8,569 of options, leaving $493 of margin before expenses and dealer spreads; caps are therefore set by rates and volatility.
- The futures remove the first-order move (index -20%: +16,734 unhedged vs +1,403 hedged; +20%: -11,873 vs +3,458) and leave the insurer long gamma and vega, so large moves and higher vol help (+1,244 for +5 vol points) while a vol fall costs. Rate P&L is mostly the bond component and is an asset-duration question. A single fitted vol likely flatters the RILA margin relative to the observed skew.

Data sources, scope, and next steps

Sources: 1y Treasury yield, observed, FRED DGS1 (fred.stlouisfed.org/series/DGS1); dividend yield, observed, Shiller/Yale (econ.yale.edu/~shiller/data.htm); 1y ATMF vol, fitted, Cboe Option EOD Summary (datashop.cboe.com/option-eod-summary); VIX cross-check, observed, Cboe VIX historical data; index level normalized to 100.

Considered: one dated indexed term per product; option-leg decomposition; bond/option/fee split with insurer sign; analytic Greeks; observed vs fitted inputs; static replication and futures delta hedge; full-revaluation stresses; rebalancing triggers.

Not considered: other IUL charges, monthly mechanics, insurance coverage, premiums, withdrawals/surrender, RILA early exit/MVA, optional benefits, loans, adviser fees, future years, real valuation and hedging costs.

Next steps: SPX skew and term structure; full Treasury and dividend curves; excluded IUL and RILA mechanics; policyholder behavior and renewals; cost-adjusted static vs dynamic comparison; gamma/vega hedges via a surface model; back-test through the October 2026 selloff.